

pins marked as +V, OP, and GND. The +V pin is given 5V supply (external), GND pin is connected to the circuit's ground, and OP pin is connected to digital pin D1 of the board. A 4.7k pull-up resistor is connected between +V and OP pin, as recommended in the datasheet for DS18B20 sensor.

The 4-digit multiplex 7-segment display is common-cathode type; its common pin is connected to ground. Its eight segment inputs (A, B, C, D, E, F, G, DP) are connected to digital pins D2 through D9, and its digit select pins 1 through 4 are connected to digital pins D10 through D13.

Arduin Nano performs almost all the tasks. It reads the temperature value from DS18B20 probe. Takes user inputs from the pot and switches. Shows water temperature on the display. Gives indications through the LEDs and buzzer. And it switches the heater on and off.

Working of the project is simple. Initially, the heater is off and red LED1 is on to indicate it. So, you may set the required water temperature using the pot and press SET push-button. You may set the temperature from 40 to 70°C, which can be seen on the 4-digit display (like 45, 48, 55, 65, etc.).

Once the required temperature is set, the heater is switched on by pressing ON push-button. Green LED2 turns on to indicate the heater is on. Now the display shows current water temperature. As the water temperature rises, the display shows the increasing water temperature. When the water temperature reaches the set temperature, the heater turns off and a beep sound is produced. The red LED turns on again.

Software

The program is written in C/C++ language using Arduino IDE. Three

libraries are used in the program: OneWire.h, DallasTemperature.h, and SevSeg.h. OneWire.h is used for the one-wire protocol to read the data using a single wire. DallasTemperature.h is used to read temperature from DS18B20 sensor. SevSeg.h is used to show temperature on the 7-segment display

The program is compiled using Arduino IDE and uploaded into internal flash memory of Arduino board's microcontroller ATmega328.

Construction and testing

After uploading the source code into internal EEPROM (flash memory) of Arduino Nano board, assemble the circuit on a general-purpose PCB. Do not forget to interconnect points 1, 2, 3, and 4 shown in Fig. 2. Before assembly, refer the author's prototype shown in Fig. 3 to know how to assemble the project. After assembling the circuit on PCB,

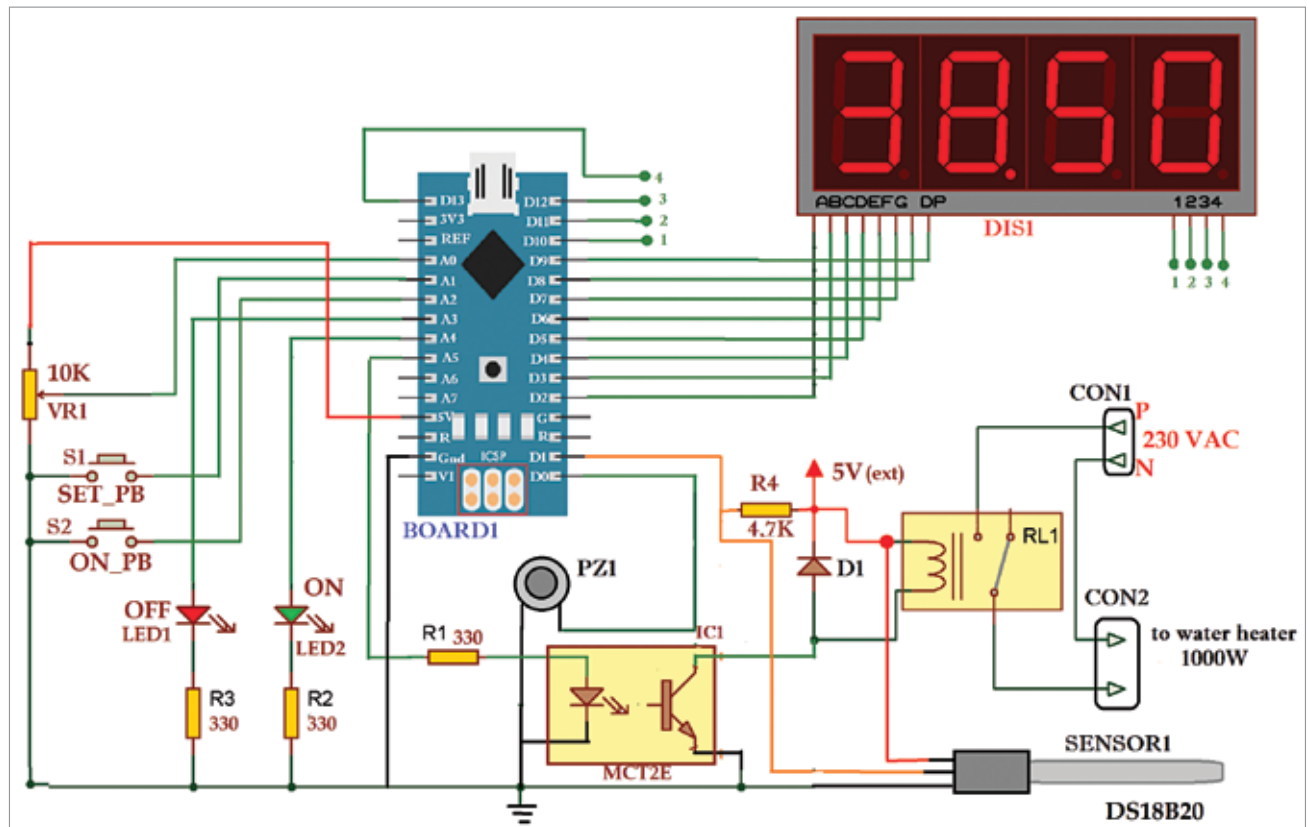


Fig. 2: Circuit diagram